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Original article

A robust multiple target tracking framework with Transformer-based data association and dynamic trajectory management in challenging environments

Xinran Zhang, Fenghua He*, Yu Yao, Zhaochen Lin

Control and Simulation Center, Harbin Institute of Technology, Harbin, 300050, China

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ABSTRACT

In this paper, a robust and reliable solution is proposed for the multiple target tracking (MTT) problem with position-only measurements in challenging environments involving unknown target numbers, high clutter, and frequent missed detections. Limited positional measurements complicate the disambiguation of closely spaced trajectories, and are further exacerbated by high false alarm and missed detection rates. To address these issues, a learning-based data association method embedded in a tracking framework with a dynamic score-based trajectory management strategy is presented. Specifically, a Cross Transformer-based Data Association (CTDA) method is proposed, which leverages cross architecture and Transformer mechanisms to effectively extract discriminative features from position measurements. Furthermore, a multiple target tracking framework is presented which integrates three key components: a Best Linear Unbiased Estimation (BLUE) filter that maintains individual trajectories while providing optimal state estimates, a dynamic trajectory scoring mechanism that continuously updates trajectory confidence based on historical association performance provided by the association network, and a trajectory management strategy that handles the full life-cycle management of trajectories, including the initialization, maintenance, and termination of the complete trajectory. Comprehensive simulation results demonstrate that the proposed method outperforms conventional algorithms in terms of tracking accuracy, reliability, and computational efficiency under complex conditions, including an over 50 % increase in data association performance and a substantial 58 % reduction in computational time.

1. Introduction

Multiple target tracking (MTT), also referred to as Multiple Object Tracking (MOT), is a critical technology employed across a wide range of applications, including surveillance [1], autonomous driving [2,3], and robotics [4,5]. It is also extensively applied in aerospace systems [6,7] as well as in sensor-based estimation [8,9]. The core challenge in MTT lies in correctly associating sensor measurements with multiple moving targets over time to estimate their trajectories and infer kinematic attributes such as velocity and acceleration. A fundamental difficulty of this challenge is the data association problem–determining the unknown correspondence between targets and measurements. This issue is further exacerbated by practical complications, including occlusions, sensor noise, clutter, and dynamic target behaviors such as targets appearing or disappearing within the scene.

Multiple target tracking methods can be broadly categorized into three classes: traditional Maximum Likelihood-based, Bayesian methods, and deep learning-based methods.

The Maximum Likelihood-type methods, including track splitting [10], joint maximum likelihood [11], 0–1 integer programming [12], and generalized correlation method [13], rely on the likelihood ratios of observed data. However, the performance of these methods often degrades in scenarios involving target intersections and dense clutter.

The conventional Bayesian-type data association methods leverage Bayesian criteria, incorporating techniques such as Nearest Neighbor (NN) [14,15], Probabilistic Data Association (PDA) [16], Joint Probabilistic Data Association (JPDA) [17], and multiple hypothesis tracking (MHT) [18,19]. While offering improved adaptability to dynamic environments (notably JPDA), conventional Bayesian methods often assume a constant number of targets. Moreover, MHT approaches face significant computational burdens as the number of tracked targets increases, limiting their use in real-time applications.

Among the Bayesian-type association methods, the Random Finite Set (RFS) framework provides a Bayesian formulation for multi-object tracking by employing conjugate priors to derive closed-form posterior distributions, thereby inherently accommodating missed detections,

* Corresponding author.
E-mail address: hefenghua@hit.edu.cn (F. He).

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Table 1
Application scopes and limitations of MTT methods.

Methods	Target number variable	False alarm	Miss detection	Fixed threshold	Calculating efficiency	Target number	Complete trajectory generation
JPDA	No	Low	No	Yes	High	Low	Yes
MHT	Yes	No/Low	Low	Yes	Low	Low	Yes
PHD	Yes	Yes	Low	Yes	High	Medium	No
CPHD	Yes	Yes	Yes	Yes	High	Medium	No
GLMB	Yes	Yes	Yes	Yes	Low	Low	Yes
DeepDA	No	No	Low	No	High	Medium	Yes
Transformer-based	Yes	Yes	Yes	No	High	Low	No
Ours	Yes	Yes	Yes	No	High	High	Yes

false alarms, and unknown numbers of targets [20–22]. Notable RFS-based filters include the Probability Hypothesis Density (PHD) filter [23,24], suitable for Poisson-distributed targets with small cardinality. However, its cardinality estimate variance increases significantly with target numbers, and it does not output target trajectories. The Cardinalized PHD (CPHD) filter relaxes the Poisson assumption but incurs superlinear computational complexity and also lacks trajectory outputs. For scenarios assuming Bernoulli distributions, the Generalized Labeled Multi-Bernoulli (GLMB) [25] improves tracking continuity and identity management. Nevertheless, all these methods require prior knowledge of the posterior density, and inaccurate approximations can lead to erroneous associations. Moreover, they share considerable computational and storage demands due to high-dimensional sampling requirements.

Recently, deep learning-based MTT algorithms have gained prominence, particularly within the domain of computer vision [26,27]. These methods train neural networks with a large number of parameters, achieving performance improvements by minimizing empirical risk on annotated datasets. These deep learning-based MTT approaches typically follow a track-by-detection paradigm [28]: targets are first detected using neural networks [29], and then associated targets based on extracted features [30]. Advanced feature extraction networks and learning-based association strategies have yielded impressive results [31–35], but they often incur increased computational overhead and complexity. Meanwhile, these approaches are highly dependent on the extraction of features from high-dimensional data, which poses limitations in scenarios where only positional data is available. Learning-based algorithms tailored for position-only MTT problems have emerged as viable alternatives, addressing some of the constraints associated with image-based methodologies. Unlike the rich, high-dimensional information provided by image-based data, limited positional measurements can complicate the differentiation of closely spaced trajectories, further challenged by high false alarm and missed detection rates [36,37]. The Long-Short-Term Memory (LSTM) networks [38] have been utilized in multiple target tracking (MTT) for simplified environments [39,40], demonstrating their capability to learn probabilistic associations from noisy radar position data and track estimates. More recently, Transformer-based architectures [41] have also been employed for data association in multiple target scenarios involving missing detections [42–44]. While significant progress has been made in improving correlation performance, current learning-based approaches still suffer from several drawbacks. For instance, [39] assumes a constant number of targets, whereas [43] and [44] rely on sequential input-output structures, making them unsuitable for complex scenarios involving trajectory generation and disappearance. Additionally, these methods remain computationally inefficient when handling extremely large numbers of targets.

As mentioned above, traditional likelihood-based or Bayesian methods are theoretically well-established, but most rely on strong model assumptions. The Random Finite Set (RFS) framework naturally supports a varying number of targets through Bayesian modeling. However, it usually requires known forms of posterior distributions, and approximation biases can easily lead to association errors. Deep learning-based methods are particularly suitable for high-dimensional data and show strong potential in feature extraction and association learning. Nevertheless, they generally depend on large amounts of annotated data, incur high computational overhead, and exhibit limited performance in position-only measurement scenarios. The application scope and limitations of various multiple target tracking methods are summarized in Table 1. A list of all abbreviations used in this study is provided in Table A.11; key mathematical symbols are summarized in Table A.12.

In this paper, we address the problem of multiple target tracking (MTT) using low-dimensional position-only measurements in challenging environments involving an unknown and time-varying number of targets, high missed detection rates, and persistent false alarms. The absence of additional discriminative features beyond position measurements creates inherent ambiguities in data association. This measurement sparsity can lead to situations where multiple distinct trajectories become observationally equivalent, significantly complicating the tracking process. To address these challenges, we propose a robust learning-based multiple target tracking framework. Specifically, we present a Cross Transformer-based Data Association (CTDA) method to effectively extract features from positional measurements, overcoming challenges of closely spaced targets and sparse data. The CTDA employs a row-column and 3-channel cross Transformer structure designed to enhance both efficiency and association accuracy. Further, this CTDA module is synergistically integrated with an MTT framework which incorporates three key components: a Best Linear Unbiased Estimation (BLUE) filter that maintains individual trajectories while providing optimal state estimates, a dynamic trajectory scoring mechanism that continuously evaluates trajectory confidence based on historical association performance provided by the CTDA network, and a robust trajectory management strategy that handles complete multiple target trajectory life cycles, including initialization, maintenance, and termination. Extensive simulations demonstrate that our proposed framework significantly outperforms existing MTT algorithms in terms of tracking accuracy, reliability, and computational efficiency under complex conditions. The key contributions of this paper are summarized as follows:

- A novel Cross Transformer-based Data Association (CTDA) method is proposed that leverages a cross Transformer mechanism to effectively extract discriminative features from position-only measurements.
- A robust and reliable MTT framework is presented that integrates the CTDA with a Best Linear Unbiased Estimation (BLUE) filter, a dynamic trajectory scoring mechanism, and a trajectory management strategy.
- A dynamic score-based trajectory management strategy is designed that evaluates trajectory confidence based on historical association performance to reliably handle the full life cycle management of trajectories for a time-varying number of targets.
- Significant performance improvements are achieved compared to established baselines under high clutter and frequent missed detections.

The remainder of this paper is organized as follows: Section 2 presents the formulation of the MTT problem. Section 3 details the cross

Transformer-based data association algorithm. Section 4 presents the trajectory scoring mechanism, the score-based trajectory management strategy, and the proposed overall MTT framework. Section 5 conducts simulation experiments to show the performance of the proposed algorithm. Finally, Section 6 concludes the paper and outlines future work.

2. Problem formulation

We consider a Multiple Target Tracking (MTT) problem under relative position-only measurement constraints. The key challenges in this scenario stem from: (1) the lack of supplementary target features makes trajectory discrimination particularly difficult, as different targets may generate identical position measurements; (2) practical sensor limitations introduce both missed detections (where valid targets go unobserved) and false alarms (where spurious measurements appear); and (3) the time-varying number of the targets, where both the quantity and identity of targets may change unpredictably over time.

Consider a group of moving targets that need to be tracked, where the number of targets may be unknown beforehand. Let $[x_i, y_i, z_i]^{\mathrm{T}} \in \mathbb{R}^3$ represent the i-th target's three-dimensional position coordinates, while v_i denotes its velocity. The target's orientation is specified by two angular parameters: ζ_i represents the pitch angle and ψ_i is the yaw angle. Choosing the state vector as $\mathbf{x}_i = [x_i, y_i, z_i, v_i, \zeta_i, \psi_i]^{\mathrm{T}}$, we have

$$\begin{aligned}\dot{\mathbf{x}}_i &= \begin{bmatrix} v_i \cos\zeta_i \cos\psi_i \\ v_i \sin\zeta_i \\ -v_i \cos\zeta_i \sin\psi_i \\ -u_{i,D} - g\sin\zeta_i \\ \frac{u_{i,L}\cos u_{i,G}}{v_i} - \frac{g}{v_i}\cos\zeta \\ -\frac{u_{i,L}\sin u_G}{v_i \cos\zeta_i} \end{bmatrix} + \boldsymbol{\eta}_i \\ &\triangleq f(\mathbf{x}_i(k-1), \boldsymbol{u}_i(k), \boldsymbol{\eta}_i(k)),\end{aligned} \tag{1}$$

where $\boldsymbol{u}_i(k) = [u_{i,D}, u_{i,L}, u_{i,G}]^{\mathrm{T}}$ are the control of velocity, pitch angle and yaw angle; η_i is the process noise vector which is assumed to be zero-mean Gaussian, i.e., $\boldsymbol{\eta}_i(k) \sim \mathcal{N}(0, \mathbf{Q}_i(k))$.

Let $r_i(k)$ denote the relative distance between the sensor and target i at time k, and $\theta_i(k)$ and $\phi_i(k)$ represent the bearing angle and elevating angle, respectively. Then the measurement model is expressed as follows

$$\begin{aligned}\mathbf{z}_i(k) &= \begin{bmatrix} r_i(k) \\ \theta_i(k) \\ \phi_i(k) \end{bmatrix} + \begin{bmatrix} \nu_{i,r}(k) \\ \nu_{i,\theta}(k) \\ \nu_{i,\phi}(k) \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{(x_i(k))^2 + (y_i(k))^2 + (z_i(k))^2} \\ \arctan\frac{y_i(k)}{x_i(k)} \\ \arctan\frac{z_i(k)}{\sqrt{x_i(k)^2 + y_i(k)^2}} \end{bmatrix} + \begin{bmatrix} \nu_{i,r}(k) \\ \nu_{i,\theta}(k) \\ \nu_{i,\phi}(k) \end{bmatrix} \\ &\triangleq h(\mathbf{x}_i(k)) + \boldsymbol{\nu}_i(k),\end{aligned} \tag{2}$$

where $\boldsymbol{\nu}_i(k) = [\nu_{i,r}(k), \nu_{i,\theta}(k), \nu_{i,\phi}(k)]^{\mathrm{T}}$ is the measurement noises assumed to be zero-mean Gaussian process, i.e., $\boldsymbol{\nu}_i(k) \sim \mathcal{N}(0, \mathbf{R}_i(k))$.

We now formulate the multiple target tracking association problem. Let $X(k) = \{\mathbf{x}_i(k) | i = 1, 2, \dots, N(k)\}$ represent the state set of all targets at the time step k, where $N(k)$ denotes the number of targets at the k-th moment. Similarly, let $Z(k) = \{\mathbf{z}_j(k) | j = 1, 2, \dots, M(k)\}$ denote the set of all measurements of the targets at time step k, where $\mathbf{z}_j(k)$ corresponds to the j-th measurement and $M(k)$ is the total number of measurements.

We define the following sets to encapsulate the temporal evolution of the system:

- $\mathcal{X}_k = \{X(k) | k = 0, 1, \dots, K\}$: the collection of all target states from the initial time step to time K,
- $\mathcal{Z}_K = \{Z(k) | k = 0, 1, \dots, K\}$: the collection of all measurements up to time K.

The multiple target tracking association problem can then be formulated as finding an association mapping:

$$\mathcal{I} : \mathcal{Z}_k \mapsto \mathcal{X}_K \tag{3}$$

such that each measurement in $\mathcal{Z}_k$ is assigned to at most one target state in $\mathcal{X}_K$, and vice versa.

However, in practice, the true states of targets are generally unknown. Consequently, the multiple target tracking problem is transformed to estimate the posterior distribution of the target states based on available measurements. Therefore, the multiple target tracking under consideration consists of data association and filtering. The data association determines whether measurements or estimates originate from the same target, while the filtering updates target states based on associated measurements. The data association module handles two primary tasks: Measurement-to-Measurement Association and Measurement-to-Track Association. The former is used for new target detection, where associations are established across multiple sampling periods, while the latter ensures track continuity by assigning new measurements to existing tracks for state updates. These associations are represented via an association matrix, which encodes the relationships between two datasets. Given two datasets: $D_1 = \{d_1^i | i = 0, 1, 2, \cdots, m\}$ and $D_2 = \{d_2^j | j = 0, 1, 2, \cdots, n\}$, the association matrix $A^{m\times n}$ is defined as

$$\mathbf{A}^{m\times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}, \tag{4}$$

where each element a_{ij} is given by

$$a_{ij} = \begin{cases} 1, & \text{if } d_1^i \text{ and } d_2^j \text{ originate from the same target;} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Note that each column and row of the association matrix may contain at most one non-zero element 1, while all other entries must be zero, ensuring one-to-one correspondence between measurements and targets.

The aim of this paper is to propose a novel multiple target tracking framework that addresses two fundamental challenges: (1) robust data association under significant measurement uncertainties and (2) accurate trajectory estimation using position-only observations. Specifically, we introduce a learning-based association mechanism that employs a neural network to compute optimal assignment probabilities, effectively resolving combinatorial ambiguities while maintaining reliable performance in challenging environments with high false alarms and missed detections. For trajectory estimation, the framework fuses noisy position measurements across multiple time steps, adaptively managing time-varying target populations, and maintaining consistent track continuity during trajectory intersection, separation, and vanishing.

3. Cross-Transformer based data association method

In this section, we present a Transformer-based data association algorithm specifically tailored for multiple target tracking scenarios involving high false alarm and missed detection rates. We first introduce a relative distance matrix to encode spatial relationships between targets and enable data normalization. Subsequently, we design the Cross-Transformer Data Association (CTDA) Network, which leverages attention mechanisms to efficiently extract and correlate features from relative distance matrices across consecutive time steps. To optimize the association performance, we further propose a specialized association matrix generation algorithm that refines and accelerates the CTDA network's output.

The overall framework of the proposed data association method is shown in Fig. 1.

3.1. Distance feature embedding and normalization

We first define a relative distance matrix to encode the spatial relationships between all possible target-measurement pairs in a structured mathematical representation and transform raw positional data into comparable relationship features. Let $D_1 = \{[x_1^i, y_1^i, z_1^i]^{\mathrm{T}} \in \mathbb{R}^3 | i = 0, 1, 2, \cdots, m\}$ and $D_2 = \{[x_2^j, y_2^j, z_2^i]^{\mathrm{T}} \in \mathbb{R}^3 | j = 0, 1, 2, \cdots, n\}$ represent two

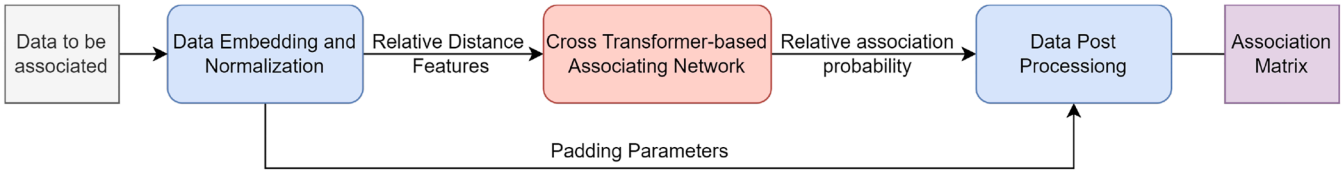


Fig. 1. Overall CTDA structure.

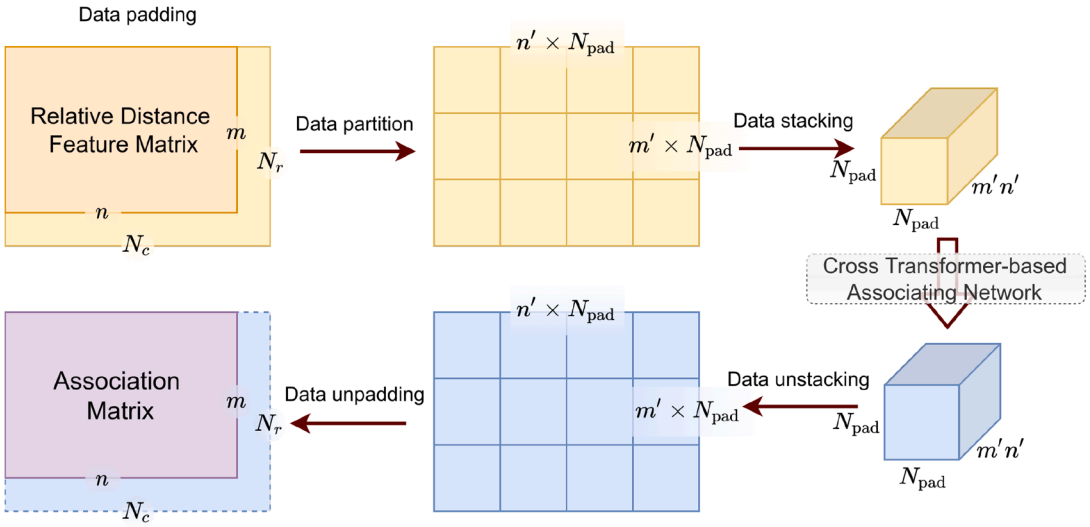


Fig. 2. Data padding and partition.

sets of 3D position data, which may include measurements, estimations, or predictions from either real targets or false alarms. To quantify the spatial relationship between D_1 and D_2, we introduce a *relative distance feature* (RDF) matrix for each spatial dimension. The RDF matrices $\mathbf{C}^x = [c_{ij}^x]_{m\times n}$, $\mathbf{C}^y = [cij^y]_{m\times n}$, and $\mathbf{C}^z = [cij^z]_{m\times n}$ capture the normalized relative distances in the x, y, and z directions respectively. The x-direction RDF matrix elements are computed as

$$c_{ij}^x = \left(\frac{2}{1+\exp\frac{-x_{\text{mid}}}{|x_2^j - x_1^i| + x_{\text{bias}}}} - 1 \right) \cdot \text{sign}(x_2^j - x_1^i), \tag{6}$$

where x_1^i and x_2^j are the x-direction data from D_1 and D_2, respectively; x_{bias} is a small quantity with appropriate size to avoid division by zero; x_{mid} represents the geometric median of relative distances, which defined as

$$x_{\text{mid}} = \sqrt{\max_{i,j}(|x_2^j - x_1^i|) \cdot (\min_{i,j}(|x_2^j - x_1^i|) + x_{\text{bias}})}. \tag{7}$$

The y and z-direction RDF matrices follow an identical construction. Notably, matrix elements approaching ± 1 indicate high similarity between corresponding data points (likely from the same target), while values near 0 suggest significant dissimilarity (likely from different targets). This inverse normalization scheme effectively transforms absolute distances into probabilistic association measures while preserving directional information through the sign function.

3.2. Dimension adaption strategy

In the MTT problem under consideration, measurement acquisition is subject to dimensional variability caused primarily by false alarms and missed detections, resulting in fluctuating data dimensions across time steps. To ensure consistent processing of these variable-dimensional measurements, we introduce a partition and padding strategy that standardizes the input dimensions while preserving the original data integrity. As illustrated in Fig. 2, this technique involves augmenting the original datasets D_1 and D_2 through strategic zero-padding. Specifically, given a predetermined partition size N_{pad}, the padding matrix dimensions is defined as $N_{\text{r}} = \lceil \frac{m}{N_{\text{pad}}} \rceil \cdot N_{\text{pad}}$ for rows and $N_{\text{c}} = \lceil \frac{n}{N_{\text{pad}}} \rceil \cdot N_{\text{pad}}$ for columns, where $\lceil \cdot \rceil$ denotes the upward rounding (ceiling) operation.

The padding matrix associated with direction x, y, and z can be constructed in a similar manner. The padding process for the x-direction RDF matrix $\mathbf{C}^x$ is implemented as:

$$\mathbf{C}_{\text{pad}}^x = \begin{bmatrix} \mathbf{C}^x & \mathbf{0}_{m\times(N_{\text{c}}-n)} \\ \mathbf{0}_{(N_{\text{r}}-m)\times n} & \mathbf{0}_{(N_{\text{r}}-m)\times(N_{\text{c}}-n)} \end{bmatrix}, \tag{8}$$

where $\mathbf{C}^x$ is defined in (6).

This zero-padding scheme maintains the original matrix characteristics by effectively introducing artificial measurements at infinite distances, equivalent to false alarms, without altering the inherent relationships in the original data. The padded matrix is subsequently partitioned into $N_{\text{pad}} \times N_{\text{pad}}$ submatrices:

$$\mathbf{C}_{\text{pad}}^x = \begin{bmatrix} \mathbf{C}_{\text{pad},11}^x & \mathbf{C}_{\text{pad},12}^x & \cdots & \mathbf{C}_{\text{pad},1n'}^x \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{C}_{\text{pad},m'1}^x & \mathbf{C}_{\text{pad},m'2}^x & \cdots & \mathbf{C}_{\text{pad},m'n'}^x \end{bmatrix}, \tag{9}$$

where each partitioned sub-matrix $C_{\text{pad},ij}^x$ is an $N_{\text{pad}} \times N_{\text{pad}}$ matrix, $m' = \lceil \frac{m}{N_{\text{pad}}} \rceil$ and $n' = \lceil \frac{n}{N_{\text{pad}}} \rceil$ represent the row and column partition counts respectively. These sub-matrices are then reshaped into a batched tensor serving as the standardized input for our cross Transformer network:

$$\mathbf{C}_{\text{bpad}}^x = [\mathbf{C}_{\text{pad},ij}^x]_{(m'\cdot n')\times N_{\text{pad}}\times N_{\text{pad}}}. \tag{10}$$

The network's output undergoes inverse processing - unstacking and unpadding - to reconstruct an association probability matrix matching the original RDF matrix dimensions, thus completing the dimensional

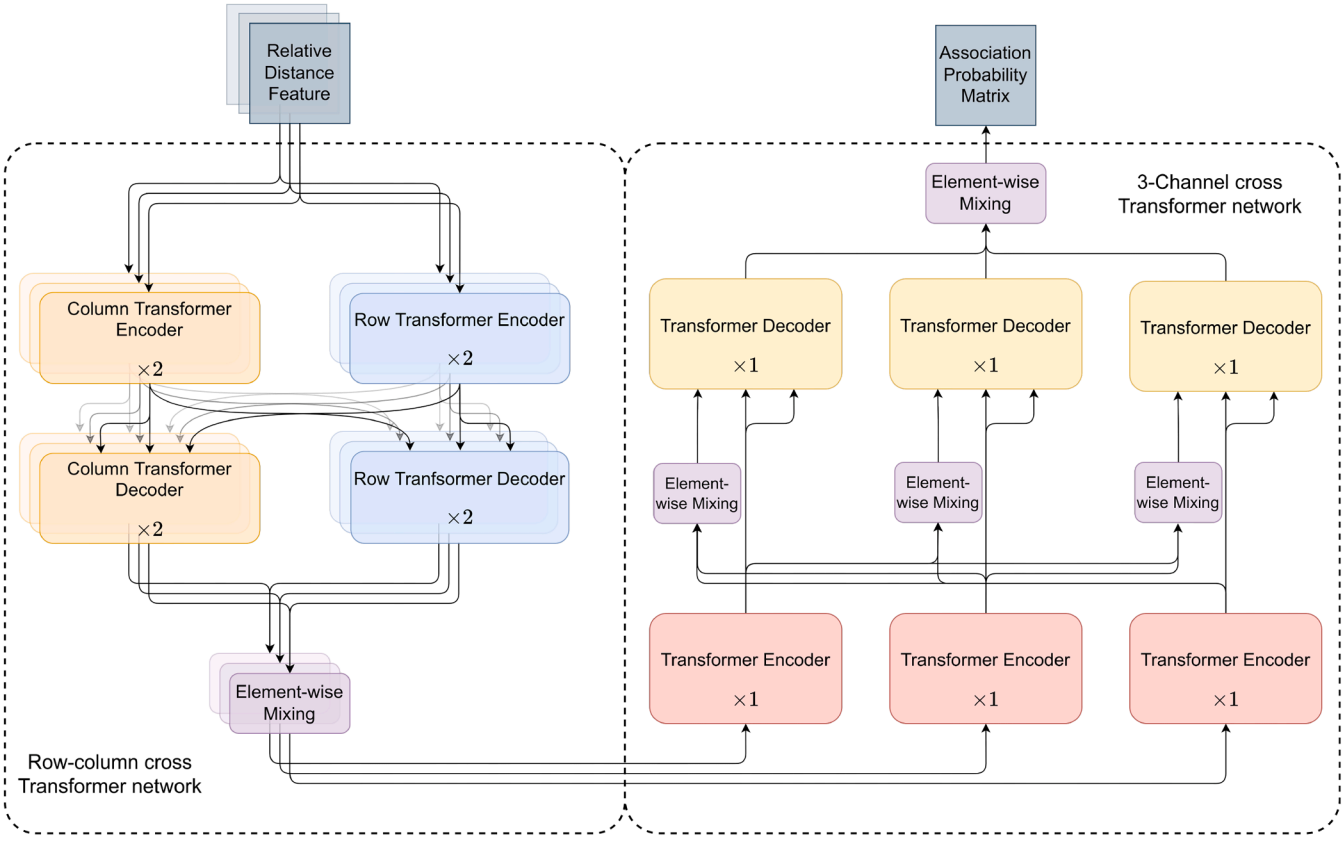


Fig. 3. Cross Transformer-based data association network.

adaptation cycle while maintaining measurement-target relationship fidelity.

3.3. Cross Transformer-based data association network design

The designed Cross Transformer-based data association network comprises two main components: the row-column cross Transformer network and the 3-channel cross Transformer network, as shown in Fig. 3.

The Transformer encoder-decoder architecture employed in our methodology presents several key modifications from the original design in [41]. Most notably, we eliminate the positional encoding module since the row and column ordering in our application carries no inherent significance.

3.3.1. Row-column cross Transformer network

As shown in Fig. 3, our row-column cross Transformer network processes three input matrix channels independently, corresponding to the spatial dimensions defined in Section 3.1. Specifically, the network architecture addresses a fundamental characteristic of RDF matrices: while rows (representing relationships from D_1 elements to D_2) and columns (representing inverse relationships) contain equivalent information, their distributions differ significantly. In the Transformer architecture, input data can only be processed one dimension at a time. To process this bidirectional relationship information without introducing dimensional bias, we propose a novel row-column cross-processing structure to separately process the rows and columns of the RDF matrix and mitigate bias arising from the processing order.

Each channel undergoes individual processing through a row Transformer encoder and a column Transformer encoder, which treat the row and column inputs as feature vectors. Each input channel is handled independently. In terms of each channel, the hidden features extracted by the encoder can be calculated by

$$\begin{cases} \mathbf{h}_{1,e} = \mathrm{ColEncoder}(\mathbf{C}_{\mathrm{bpad}}), \\ \mathbf{h}_{2,e} = \mathrm{RowEncoder}(\mathbf{C}_{\mathrm{bpad}}{}^{\mathrm{T}})^{\mathrm{T}}. \end{cases} \tag{11}$$

The decoder architecture employs a sophisticated cross-feeding mechanism where: Encoder outputs serve as Key/Value inputs for the second MHA module, and both encoder outputs are mutually exchanged as inputs to the counterpart decoder's first MHA module. Specifically, the outputs from the encoders $\mathbf{h}_{1,e}$ and $\mathbf{h}_{2,e}$ for each channel are fed into row and column Transformer decoders in a cross pattern. The Transformer decoder consists of two multi-head attention (MHA) modules. In the row-column cross Transformer network, the outputs of the row and column encoders serve as the *Key* and *Value* inputs for the second MHA module, respectively. Meanwhile, the outputs of each encoder are cross-fed into the other decoder, serving as inputs to the layer of the first MHA module. The extracted feature of the column decoder can be calculated by

$$\begin{cases} \mathbf{h}_{1,d_1} = \mathrm{LN}(\mathrm{MHA}(\mathbf{h}_{1,e}, \mathbf{h}_{1,e}, \mathbf{h}_{1,e}) + \mathbf{h}_{1,e}), \\ \mathbf{h}_{1,d_2} = \mathrm{LN}(\mathrm{MHA}(\mathrm{FFN}(\mathbf{h}_{2,e}^{\mathrm{T}})^{\mathrm{T}}, \mathbf{h}_{1,d_1}, \mathbf{h}_{1,d_1}) + \mathbf{h}_{1,d_1}), \\ \mathbf{h}_{1,d} = \mathrm{LN}(\mathrm{FFN}(\mathbf{h}_{1,d_2}) + \mathbf{h}_{1,d_2}), \end{cases} \tag{12}$$

where $\mathrm{LN}(\cdot)$ denotes the layer normalization, $\mathrm{MHA}(Query, Key, Value)$ denotes the multi-head attention layer with input $Query$, Key and $Value$, FFN denote the feed forward network. In the second MHA layer, the encoded feature $\mathbf{h}_{2,e}$ is fed into a feed forward layer to match the dimension of $\mathbf{h}_{1,d_1}$ and $\mathbf{h}_{2,e}$.

Similarly, the feature of the row decoder can be calculated by

$$\begin{cases} \mathbf{h}_{2,d_1} = \mathrm{LN}(\mathrm{MHA}(\mathbf{h}_{2,e}^{\mathrm{T}}, \mathbf{h}_{2,e}^{\mathrm{T}}, \mathbf{h}_{2,e}^{\mathrm{T}}) + \mathbf{h}_{2,e}^{\mathrm{T}})^{\mathrm{T}}, \\ \mathbf{h}_{2,d_2} = \mathrm{LN}(\mathrm{MHA}(\mathrm{FFN}(\mathbf{h}_{1,e})^{\mathrm{T}}, \mathbf{h}_{2,d_1}^{\mathrm{T}}, \mathbf{h}_{2,d_1}^{\mathrm{T}}) + \mathbf{h}_{2,d_1}^{\mathrm{T}})^{\mathrm{T}}, \\ \mathbf{h}_{2,d} = \mathrm{LN}(\mathrm{FFN}(\mathbf{h}_{2,d_2}^{\mathrm{T}}) + \mathbf{h}_{2,d_2}^{\mathrm{T}})^{\mathrm{T}}. \end{cases} \tag{13}$$

The outputs of the row and column decoders, $\mathbf{h}_{1,d}$ and $\mathbf{h}_{2,d}$, which consist of two matrices for each channel, are combined using a mixing module.

To accommodate diverse requirements, we introduce two mixing strategies:

- Element-wise mixing (computationally efficient): the process is

$$\begin{aligned} \mathbf{h}_{\mathrm{d}} = \tanh(&\mathrm{Reshape}(\mathbf{W}^{2\times 1}\mathrm{Reshape}([\mathbf{h}_{1,d}; \mathbf{h}_{2,d}], \\ &N_{\mathrm{pad}}, N_{\mathrm{pad}}, 2) + \mathbf{b}, N_{\mathrm{pad}}, N_{\mathrm{pad}})), \end{aligned} \tag{14}$$

  where $\mathbf{h}_{\mathrm{d}}$ is the output of the mixed matrix, $[\cdot]$ denotes the concatenate operation, $\mathrm{Reshape}(\cdot)$ represents converting the dimension of the matrix to a prescribed number, $\mathbf{W}$ and $\mathbf{b}$ are trainable weights and biases, $\mathbf{h}_1$ and $\mathbf{h}_2$ are outputs of the decoders. For the sake of simplicity and clarity, the matrices $\mathbf{h}_{1,d}$, $\mathbf{h}_{1,d}$, and $\mathbf{h}^{\mathrm{d}}$ are assumed to be two-dimensional square matrices, with the batch size omitted.
- Fully-connected mixing (higher accuracy): the process is

$$\begin{aligned} \mathbf{h}_{\mathrm{d}} = \tanh(&\mathrm{Reshape}(\mathbf{W}^{2\times N_{\mathrm{pad}}\times N_{\mathrm{pad}}}\mathrm{Reshape}([\mathbf{h}_{1,d}; \mathbf{h}_{2,d}], \\ &2\cdot N_{\mathrm{pad}}\cdot N_{\mathrm{pad}}, 1) + \mathbf{b}, N_{\mathrm{pad}}, N_{\mathrm{pad}})). \end{aligned} \tag{15}$$

While the fully-connected mixing approach achieves stronger representational capacity by preserving more comprehensive feature interactions, its computational demands make it impractical for real-time applications. The element-wise alternative, though potentially less expressive, offers superior computational efficiency due to its simpler architecture and fewer parameters. Our subsequent experimental results confirm that this architectural choice delivers the best compromise between accuracy and computational performance for target tracking applications.

3.3.2. 3-channel cross Transformer network

The three-channel RDF matrix, designated as the x, y, and z channels, undergoes feature matrix transformation through our row-column cross Transformer structure to extract a comprehensive spatial relationship features between datasets D_1 and D_2. To enable effective cross-channel information fusion while maintaining spatial symmetry, we introduce a novel 3-channel cross Transformer structure. The processing pipeline begins with three parallel Transformer encoders operating on the element-wise mixed outputs. The encoding process can be defined as

$$\begin{cases} \mathbf{h}_{x,e} = C_x\mathrm{Encoder}(\mathbf{h}_{x,d}), \\ \mathbf{h}_{y,e} = C_y\mathrm{Encoder}(\mathbf{h}_{y,d}), \\ \mathbf{h}_{z,e} = C_z\mathrm{Encoder}(\mathbf{h}_{z,d}), \end{cases} \tag{16}$$

where $\mathbf{h}_{x,e}$, $\mathbf{h}_{x,e}$, and $\mathbf{h}_{x,e}$ represent the hidden feature of each spatial dimension.

Subsequently, the output from each encoder, corresponding to each channel, is simultaneously fed into the first MHA module of the decoder for the other channels. Furthermore, each output is concurrently employed as both the *key* and *value* inputs for the second Multi-Head Attention (MHA) module in the respective channel's decoder, while the aggregated features from the other two channels constitute the *query*. To illustrate this process, the decoding for the x-channel is formulated as follows:

$$\begin{cases} \mathbf{h}_{x,d_1} = \mathrm{LN}(\mathrm{MHA}(\mathbf{h}_{x,e}, \mathbf{h}_{x,e}, \mathbf{h}_{x,e}) + \mathbf{h}_{x,e}), \\ \mathbf{h}_{x,d_q} = \mathrm{ElementMix}(\mathbf{h}_{y,e}, \mathbf{h}_{z,e}), \\ \mathbf{h}_{x,d_2} = \mathrm{LN}(\mathrm{MHA}(\mathbf{h}_{x,d_q}, \mathbf{h}_{x,d_1}, \mathbf{h}_{x,d_1}), \\ \mathbf{h}_{x,d} = \mathrm{LN}(\mathrm{FFN}(\mathbf{h}_{x,d_2}), \end{cases} \tag{17}$$

where $\mathrm{ElementMix}(\cdot,\cdot)$ denotes the element-wise mixing process.

The final probability output is computed through the element-wise mixing operations:

$$\begin{aligned} \mathbf{O}^{\mathrm{pr}} = &\mathrm{sigmoid}(\mathrm{Reshape}(\mathbf{W}^{3\times 1}\mathrm{Reshape}([\mathbf{h}_{x,d}; \mathbf{h}_{y,d}; \mathbf{h}_{z,d}], \\ &N_{\mathrm{pad}}, N_{\mathrm{pad}}, 3) + \mathbf{b}, N_{\mathrm{pad}}, N_{\mathrm{pad}})), \end{aligned} \tag{18}$$

where $\mathbf{h}_{x,d}$, $\mathbf{h}_{y,d}$ and $\mathbf{h}_{z,d}$ are outputs of the decoders corresponding to three channels.

Notably, the cross Transformer employed in the 3-channel cross Transformer module differs from its counterpart in the row-column cross Transformer. Specifically, the 3-channel cross Transformer module comprises a reduced number of layers in comparison to the row-column Transformer module. This adjustment is made by the observation that the fusion of information across distinct channels tends to be comparatively less intricate than the operations entailed by the row-column cross Transformer module. Therefore, to avoid overfitting, we do not employ a deeper network within this particular context. Additionally, the number of Transformers in the crossing structure applied between each pair of channels is not fixed, which flexibly varies with the dimension of the data. The number of encoders and decoders may be different. This design achieves efficient cross-channel information fusion while maintaining computational tractability and model generalization capability.

3.3.3. Loss function

The data association matrix is inherently sparse, represented as a binary (0–1) matrix where the number of true associations (1s) is significantly smaller than either the matrix dimensions or the number of non-associations (0s). However, these rare positive elements carry substantially greater importance than the predominant negative elements, as they represent the critical matching pairs that fundamentally determine association accuracy. This imbalance poses two key challenges: (1) the difficulty in effectively learning these informative but sparse positive associations, and (2) the disproportionate influence of the more numerous but less informative negative samples during training.

To address these challenges, we employ the Focal Loss [45] as our optimization objective, which strategically reweights the loss contribution of each element based on its classification difficulty and class imbalance:

$$\begin{aligned} L = \sum_{i,j\le N_{\max}} &-\alpha_t O_{ij}^{\mathrm{gt}}\cdot(1-O_{ij}^{\mathrm{pr}})^{\gamma_t}\log O_{ij}^{\mathrm{pr}} \\ &-(1-\alpha_t)(1-O_{ij}^{\mathrm{gt}})(O_{ij}^{\mathrm{pr}})^{\gamma_t}\log(1-O_{ij}^{\mathrm{pr}}), \end{aligned} \tag{19}$$

where O_{ij}^{pr} and O_{ij}^{gt} represent the elements at i-th row and j-th column of the predicted association probability matrix $\mathbf{O}^{\mathrm{pr}}$ and the ground truth association matrix $\mathbf{O}^{\mathrm{gt}}$, respectively; $\lambda_t \ge 0$ is a tunable focusing parameter, and $0 \le \alpha_t \le 1$ is a balance factor. Through empirical validation, $\lambda_t = 2$ and $\alpha_t = 0.75$ are utilized to optimally handle the extreme class imbalance while maintaining effective learning of both positive and negative associations.

3.4. Association matrix generation

Each element in the association probability matrix $\mathbf{O}^{\mathrm{pr}}$, generated by our Transformer-based data association network, quantifies the likelihood of association between a given data pair. The network output requires further processing to produce the association (0–1) matrix.

In practice, a threshold-based binarization typically suffices for most scenarios:

$$A_{ij} = \begin{cases} 1 & \textbf{if} O_{ij}^{pr} \ge \lambda_a, \\ 0 & \textbf{otherwise}, \end{cases} \tag{20}$$

where λ_a represents the association threshold and A_{ij} denotes the (i,j)-th element of the association matrix $\mathbf{A}$.

However, this threshold-based approach may yield ambiguous results in complex scenarios like intersecting target trajectories. To ensure robust association matrix generation across all cases, we integrate

the Kuhn-Munkres (KM) algorithm [30] for optimal assignment. See Algorithm 1 for implementation details.

Algorithm 1 Network result processing.

Input: Output of the Transformer Network $\mathbf{O}^{pr}$; Association Threshold λ_a ;
Output: Association Matrix $\mathbf{A}$;
Initiate Mask $\mathbf{M} = \mathbf{1}^{m\times n}$, $\mathbf{A} \Leftarrow \mathbf{0}^{m\times n}$;
$\mathbf{M}[\mathbf{O} < \lambda_a] \Leftarrow 0$;
for $i = 1{:}\mathrm{m}$ **do**
  for $j = 1{:}\mathrm{n}$ **do**
    if $\mathbf{A}[-i, j] == 0$ *and* $\mathbf{A}[i, -j] == 0$ **then**
      $\mathbf{M}[i, :], \mathbf{M}[:, j] \Leftarrow 0$;
      $\mathbf{A}[i, j] \Leftarrow 1$;
    end if
  end for
end for
rowIdx,colIdx = KMLinearAssign($-\mathbf{A}[\mathbf{M} == 1]$);
$\mathbf{A}$[rowIdx,colIdx] $\Leftarrow 1$;

4. Overall multiple target tracking framework

In this section, we first establish the correspondence between the predicted trajectories and the current measurements. Each trajectory is independently maintained using a Best Linear Unbiased Estimation (BLUE) filter to ensure optimal state estimation. We then introduce a trajectory scoring mechanism that dynamically updates each trajectory's confidence score based on its historical association performance. To obtain reliable multiple target tracking, we propose a comprehensive trajectory management strategy that includes: new trajectory initiation for potential targets, existing trajectory confirmation and maintenance, and obsolete trajectory termination. Finally, we present the overall multiple target tracking framework and algorithm.

4.1. Trajectory-measurement correlation analysis

We first establish the relevance between the predicted trajectories and the current measurements. Consider p predicted position trajectories, denoted as $\{\hat{\mathbf{x}}_i(k|k-1)|i = 1, 2, \cdots, p\}$, and q measurements, represented as $\{\mathbf{z}_j(k)|j = 1, 2, \cdots, q\}$, at the time instant k. Each trajectory is maintained by a Best Linear Unbiased Estimation (BLUE) filter, which will be introduced subsequently.

Let $\mathbf{A}(k) \in \mathbb{R}^{p\times q}$ denote the association matrix provided by the CTDA algorithm at time k. Note that $\mathbf{A}(k)$ is typically not of full rank due to missed detection and false alarms in the measurements. We denote the (i, j)-th element in $\mathbf{A}(k)$ as $a_{ij}(k), i \in \{1, 2, \ldots, p\}$, $j \in \{1, 2, \ldots, q\}$, and $\mathbf{r}_i$ and $\mathbf{c}_j$ represent its i-th row and j-th column vectors of $\mathbf{A}(k)$, respectively. We assume that each measurement or prediction belongs to one of the following three cases:

- Measurement without an associated trajectory: $\{\mathbf{z}_j(k)|\sum_{i=1}^{p}\mathbf{c}_j = 0\}$,
- Trajectory without an associated measurement: $\{\hat{\mathbf{x}}_i(k|k-1)|\sum_{j=1}^{q}\mathbf{r}_i = 0\}$,
- Associated measurement-trajectory pairs: $\{(\hat{\mathbf{x}}_i(k|k-1), \mathbf{z}_j(k))|\sum\mathbf{r}_i = \sum\mathbf{c}_j = a_{ij} = 1\}$.

4.2. Best linear unbiased estimation

Each trajectory is independently maintained using a filter. Consider the motion and measurement models for each target, as described in (1) and (2), which are nonlinear. Nonlinear estimation methods, such as the Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), or particle filter, can be employed for state estimation.

In the problem under consideration, the measurements are constrained to relative distance r, bearing θ, and elevation ϕ with respect to the target. For such scenario, a specialized variant of the Extended Kalman filter, known as the Best Linear Unbiased Estimating Filter (BLUE) [46], is particularly suitable to obtain optimal estimates in terms of accuracy and efficiency.

We next perform the BLUE filtering to obtain the predicted and estimated trajectories for the MTT problem. For notation brevity, we define $\check{r} = \sqrt{\hat{x}(k|k-1)^2 + \hat{y}(k|k-1)^2 + \hat{z}(k|k-1)^2}$, and $\check{r}_1 = \sqrt{\hat{x}(k|k-1)^2 + \hat{y}(k|k-1)^2}$. Additionally, let $\lambda_1 = E[\cos\tilde{\theta}] = e^{-\sigma_\theta^2/2}$, $\lambda_2 = E[\cos^2\tilde{\theta}] = \frac{1}{2}(1 + e^{-2\sigma_\theta^2})$, $\lambda_3 = E[\sin^2\tilde{\theta}] = \frac{1}{2}(1 - e^{-2\sigma_\theta^2})$, $\mu_1 = E[\cos\tilde{\phi}] = e^{-\sigma_\phi^2/2}$, $\mu_2 = E[\cos^2\tilde{\phi}] = \frac{1}{2}(1 + e^{-2\sigma_\phi^2})$ and $\mu_3 = E[\sin^2\tilde{\phi}] = \frac{1}{2}(1 - e^{-2\sigma_\phi^2})$ and denote $\alpha = \left(\frac{\mu_2\sigma_r^2}{\check{r}^2} + \frac{\mu_3\hat{z}(k|k-1)^2}{\check{r}_1^2} + \frac{\mu_3\sigma_r^2\hat{z}(k|k-1)^2}{\check{r}_1^2\check{r}^2}\right)$, $\alpha_1 = (\lambda_2\mu_2 - \lambda_1^2\mu_1^2)\hat{x}(k|k-1)^2 + \lambda_3\mu_2\hat{y}(k|k-1)^2$, $\alpha_2 = (\lambda_2\mu_2 - \lambda_1^2\mu_1^2)\hat{y}(k|k-1)^2 + \lambda_3\mu_2\hat{x}(k|k-1)^2$, $\alpha_3 = (\mu_2 - \mu_1^2)\hat{z}(k|k-1)^2 + \mu_3(\hat{x}(k|k-1)^2 + \hat{y}(k|k-1)^2)$, $\alpha_4 = (\mu_2(\lambda_2 - \lambda_3) - \lambda_1^2\mu_1^2)\hat{x}(k|k-1)\hat{y}(k|k-1)$ and $\alpha_5 = (\lambda_1(\mu_2 - \mu_3) - \lambda_1^2\mu_1^2)\hat{z}(k|k-1)$.

Propagation: Let $\hat{\mathbf{x}}(k-1|k-1)$ and $\mathbf{P}(k-1|k-1)$ be the state estimation and covariance at time instant $k-1$. Let $\mathbf{P}(k|k-1)$ and $\mathbf{P}(k|k)$ denote the prediction and the estimate of the covariance matrix at time k, and $\hat{\mathbf{x}}(k|k-1)$ and $\hat{\mathbf{x}}(k|k)$ denote the prediction and the estimate of the states at time k. Then the propagation for the state and the covariance matrix are given by

$$\hat{\mathbf{x}}(k|k-1) = \mathbf{F}(k)\hat{\mathbf{x}}(k-1|k-1) + \mathbf{G}(k)\boldsymbol{\eta}(k) \tag{21}$$

and

$$\mathbf{P}(k|k-1) = \mathbf{F}(k)\mathbf{P}(k-1|k-1)\mathbf{F}(k)^{\mathrm{T}} + \mathbf{G}(k)\mathbf{Q}_k\mathbf{G}(k)^{\mathrm{T}}, \tag{22}$$

where $\mathbf{F}(k) = \frac{\partial f}{\partial\mathbf{x}(k-1)}\Big|_{\hat{\mathbf{x}}(k-1|k-1)}$, $\mathbf{G}(k) = \frac{\partial f}{\partial\boldsymbol{\eta}(k)}\Big|_{\hat{\mathbf{x}}(k-1|k-1)}$ and $\mathbf{Q}_k$ denote the covariance matrix of the process noise.

Update: To track multiple targets simultaneously, a family of (BLUE) filters associated with trajectories is employed to obtain the trajectory estimates of the targets. The update procedures of the state and covariance are

$$\hat{\mathbf{x}}(k|k) = \hat{\mathbf{x}}(k|k-1) + \mathbf{K}(k)(\mathbf{z}_k - \mu_1[\lambda_1\hat{x}(k|k-1), \lambda_1\hat{y}(k|k-1), \hat{z}(k|k-1)]^{\mathrm{T}}), \tag{23}$$

and

$$\mathbf{P}(k|k) = (\mathbf{I} - \mathbf{K}(k)\mathbf{H}(k))\mathbf{P}(k|k-1), \tag{24}$$

where $\mathbf{I}$ is an identity matrix, $\mathbf{H}(k) = \frac{\partial h}{\partial\mathbf{x}(k)}\Big|_{\hat{\mathbf{x}}(k|k-1)}$.

The Kalman gain matrix $\mathbf{K}(k)$ is calculated by

$$\mathbf{K}(k) = \mu_1\mathbf{P}(k|k)\mathbf{J}\mathbf{S}^{-1}, \tag{25}$$

where $\mathbf{J}$ is a transforming matrix that operates $\mathbf{P}(k|k)\mathbf{J}$ to combine the columns of $\mathbf{P}(k|k)$ corresponding to x, y, and z into a new $n\times 3$ matrix, and the elements of symmetric matrix $\mathbf{S} = [s_{ij}]_{3\times 3}$ are given by:

$$s_{11} \approx \lambda_2\mu_2\Sigma_x + \lambda_3\mu_2\Sigma_y + \alpha(\lambda_2\hat{x}(k|k-1)^2 + \lambda_3\hat{y}(k|k-1)^2) + \alpha_1,$$
$$s_{22} \approx \lambda_2\mu_2\Sigma_y + \lambda_3\mu_2\Sigma_x + \alpha(\lambda_3\hat{x}(k|k-1)^2 + \lambda_2\hat{y}(k|k-1)^2) + \alpha_2,$$
$$s_{33} \approx \mu_2\Sigma_z + \mu_3\left(\Sigma_x + \Sigma_y + \mu_2\sigma_r^2\frac{\hat{z}(k|k-1)^2}{\check{r}^2} + \mu_3\sigma_r^2\frac{\check{r}_1^2}{\check{r}^2}\right) + \alpha_3,$$
$$s_{12} = s_{21} \approx (\lambda_2 - \lambda_3)(\mu_2\Sigma_{xy} + \alpha\hat{x}(k|k-1)\hat{y}(k|k-1)) + \alpha_4,$$
$$s_{13} = s_{31} \approx \lambda_1(\mu_2 - \mu_3)\left(\Sigma_{xz} + \sigma_r^2\frac{\hat{x}(k|k-1)\hat{z}(k|k-1)}{\check{r}^2}\right) + \alpha_5\hat{x}(k|k-1),$$
$$s_{23} = s_{32} \approx \lambda_1(\mu_2 - \mu_3)\left(\Sigma_{yz} + \sigma_r^2\frac{\hat{y}(k|k-1)\hat{z}(k|k-1)}{\check{r}^2}\right) + \alpha_5\hat{y}(k|k-1),$$

with Σ_x, Σ_y and Σ_z being the corresponding estimation variances, and Σ_{xy}, Σ_{yz} and Σ_{xy} beging the covariance between the estimations.

4.3. Trajectory confidence scoring mechanism

We propose a probabilistic trajectory confidence scoring mechanism that dynamically evaluates each trajectory's reliability based on its historical association performance. The mechanism operates within the following framework: Let $\mathbb{L} = \{i | i \in \mathbb{N}^*\}$ denote a universal label set for all potential trajectories with$\mathbb{N}^*$ being the set of positive integers. Then $\mathbb{L}$ can be partitioned into two disjoint subsets $\mathbb{L}^{\mathrm{m}} \cap \mathbb{L}^{\mathrm{c}} = \emptyset$ with:

- $\mathbb{L}^{\mathrm{m}}$: Labels for maintained trajectories (confirmed targets),
- $\mathbb{L}^{\mathrm{c}}$: Labels for candidate trajectories (unconfirmed targets).

Note that candidate trajectories are promoted to maintained status only after achieving sufficient evidence.

To take into account the trajectory's historical association information, we introduce a confidence score $S_l(k)$ for trajectory $l \in (\mathbb{L}^{\mathrm{m}} \cup \mathbb{L}^{\mathrm{c}})$ at time instant k evolving through a score dynamics with truncated and attenuated forms as follows:

$$S_l(k) = \lambda_s S_l(k-1) + \Delta S_l(k) - (\lambda_s)^{\lambda_t} \Delta S_l(k-\lambda_t), \tag{26}$$

where $\lambda_s \in (0, 1]$ is a decay factor (older associations contribute less), $\lambda_t \in \mathbb{N}^*$ is a temporal window size for score truncation, and $\Delta S_l(k)$ represents the increment of trajectory score at moment k, which can also be considered as the evaluation score of the relation between the measurement and trajectory with label l.

The confidence score evolution in (26) ensures that recent associations ($\Delta S_l(k)$) dominate, long-term memory is preserved but attenuated, and outdated contributions ($k - \lambda_t$) are actively forgotten. The key in (26) is the design of the score increment. We design the instantaneous score increment $\Delta S_l(k)$ as the following three cases:

- New candidate initialization: For a measurement $\mathbf{z}_k^j$ that lacks associations, a new candidate trajectory l is initialized by a filter with a new label added into $\mathbb{L}^{\mathrm{c}}$. The state of the target is initialized with measurements and priors. Let t_l^{b} be the initialization time for the tracked trajectory with label l. We assume that $S_l(k) = 0$ for $k \le t_l^{\mathrm{b}}$. An initial score of the trajectory l is given in birth/clutter likelihood ratio (LLR) with log form by [47]:

  $$\Delta S_l(k) = \ln \frac{P_{\mathrm{B}}}{P_{\mathrm{FA}}}, \tag{27}$$

  where P_{B} denotes the target birth probability and P_{FA} represents the probability that a clutter point cannot be distinguished from a true target (false alarm rate). Both P_{B} and P_{FA} can be pre-calculated based on sensor characteristics.
- Missed detection: For unassociated trajectory l, the state updates rely purely on prediction,

  $$\hat{\mathbf{x}}_l(k|k) = \hat{\mathbf{x}}_l(k|k-1), \tag{28}$$

  and

  $$\hat{P}_l(k|k) = \hat{P}_l(k|k-1). \tag{29}$$

  The increment of the corresponding trajectory score is given by:

  $$\Delta S_l(k) = \ln \frac{1 - P_{\mathrm{FA}}}{1 - P_{\mathrm{D}}}, \tag{30}$$

  where P_{D} denotes the sensor detection probability.
- Successful association: For trajectory-measurement pairs, i.e., a trajectory l associated with a measurement $\mathbf{z}_j(k)$, the BLUE filter updates described in (23) and (24) yield the state and covariance estimates for trajectory l. The increment of the corresponding trajectory score can be calculated based on the position measurement and the estimation of the trajectory l,

  $$\Delta S_l(k) = \ln \mathrm{pr}(\hat{\mathbf{x}}_l(k|k) | \hat{\mathbf{x}}_l(k|k-1)) - \ln P_{\mathrm{FA}}, \tag{31}$$

  where $\mathrm{pr}(\cdot)$ denotes the probability density function, for simplicity we assume that the probability density function of $\hat{\mathbf{x}}(k|k)$ is subject to a Gaussian distribution $\mathcal{N}(\hat{\mathbf{x}}_l(k|k-1), \hat{P}_l(k|k-1))$,

  $$\mathrm{pr}(\hat{\mathbf{x}}_l(k|k) | \hat{\mathbf{x}}_l(k|k-1)) = \frac{1}{2\pi\sqrt{|\hat{P}_l(k|k-1)|}} \exp\left(-\frac{1}{2} d_l^2\right), \tag{32}$$

  where $d_l^2 = (\hat{\mathbf{x}}_l(k|k) - \hat{\mathbf{x}}_l(k|k-1))^{\mathrm{T}} \hat{P}_l(k|k-1)^{-1} (\hat{\mathbf{x}}_l(k|k) - \hat{\mathbf{x}}_l(k|k-1))$.

  Additionally, assume that the clutter spatial distribution is uniformly distributed over the area D and the number of clutter N_{FA} obeys a Poisson distribution with parameter λ_c as follows

  $$\Pr(N_{\mathrm{FA}} = i) = \frac{\lambda_c^i}{i!} e^{-\lambda_c}, \tag{33}$$

  where Pr denotes the probability.

  Substituting (32) and (33) into (31) yields

  $$\begin{aligned} \Delta S_l(k) &= \ln \mathrm{pr}(\hat{\mathbf{x}}_l(k|k) | \hat{\mathbf{x}}_l(k|k-1)) - \ln P_{\mathrm{FA}} \\ &= \ln \frac{1}{2\pi\sqrt{|\hat{P}_l(k|k-1)|}} e^{-\frac{1}{2} d_l^2} - \ln \frac{1}{D} \Sigma_{i=1}^{\infty} \frac{\lambda_c^i}{i!} e^{-\lambda_c} \\ &= \ln \frac{D}{2\pi\lambda_c} - \frac{1}{2} \ln |\Sigma_k^l| - \frac{1}{2} d_l^2. \end{aligned} \tag{34}$$

Remark 1. Although parameters, such as P_A, P_B and P_{FA}, are inherently scene-dependent, subsequent experimental results show that robust performance can be achieved by adjusting the weighting factors λ_s and λ_t within reasonable ranges for most scenarios. This implies that in practical implementations, exact values for these scene-dependent parameters are not strictly necessary; approximations that are reasonably close to their true values typically suffice for maintaining effective tracking performance.

4.4. Score-based trajectory management

To obtain reliable results in tracking an unknown and time-varying number of targets, we propose a trajectory management strategy. This strategy dynamically handles trajectory initialization, maintenance, and termination based on accumulated evidence from historical measurements. A predefined confirmation threshold $S^{\mathrm{c}} \in \mathbb{R}$ is defined to distinguish tentative detections from confirmed tracks.

- Candidate trajectories ($l \in \mathbb{L}^{\mathrm{c}}$): For unconfirmed trajectories, the management rules are:
  - If $S_l(k) \ge S^{\mathrm{c}}$, the trajectory l is sufficiently confirmed. Its label is transferred from $\mathbb{L}^{\mathrm{c}}$ and added to $\mathbb{L}^{\mathrm{m}}$, making it as a persistent target.
  - If $0 \le S_l(k) < S^{\mathrm{c}}$, the trajectory l remains in $\mathbb{L}^{\mathrm{c}}$ for further evidence accumulation. This hysteresis prevents premature confirmation of noisy detections.
  - If $S_l(k) < 0$, the trajectory is deemed unreliable and is deleted along with its associated filter. This occurs when consistent misassociations outweigh positive evidence.
- Maintained trajectories ($l \in \mathbb{L}^{\mathrm{m}}$): For confirmed trajectories:
  - if $S_l(k) \ge 0$, the trajectory continues to be tracked. The positive score indicates sustained measurement associations.
  - Otherwise if $S_l(k) < 0$, the trajectory is terminated and its label is retired (marked with its current label). This occurs when the target exits the surveillance area or tracking fails irrecoverably.

The confirmation threshold S^{c} is typically set empirically based on the expected signal-to-clutter ratio, where higher values reduce false confirmations at the cost of increased latency in detecting new targets.

4.5. Overall multiple target tracking framework

The overall framework of the proposed multiple target tracking method is shown in Fig. 4.

The framework consists of three key components: (1) a learning-based measurement-to-trajectory association module that establishes

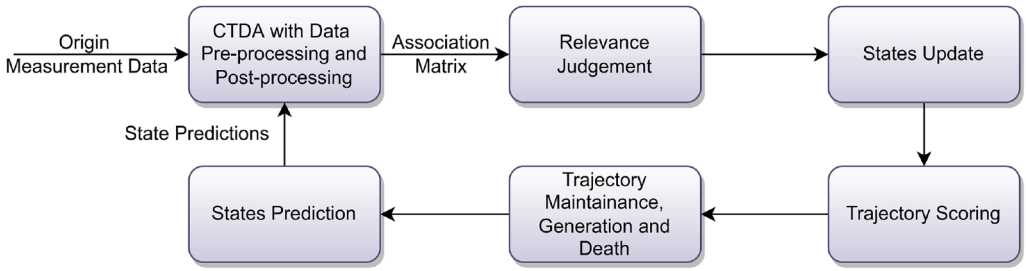


Fig. 4. Multiple target tracking framework.

correspondence between predictions and observations, where each trajectory is independently maintained using a Best Linear Unbiased Estimation (BLUE) filter for optimal state estimation; (2) a probabilistic scoring mechanism that dynamically evaluates trajectory confidence by accumulating historical association evidence; and (3) an adaptive management strategy that automatically handles trajectory lifecycle operations - including new trajectory initiation for potential targets, confirmation of reliable tracks, maintenance of established trajectories, and termination of obsolete ones. By integrating these components into a unified architecture, the proposed framework achieves robust tracking performance while maintaining computational efficiency, as demonstrated in subsequent experimental validation. The pseudo code is given in Algorithm 2.

Algorithm 2 Multiple target tracking algorithm.

Input: Measurements at each time step, $\mathcal{Z} = \{\mathbf{z}(k)|k \in T\}$; Time step, $T = [0, 1, 2, \cdots, t_{\text{end}}]$;
Output: The estimated trajectories at each time step, $\hat{X}(k|k)$;

Initialize maintaining label set $\mathbb{L}^{\text{m}} = \emptyset$, candidate label set $\mathbb{L}^{\text{c}} = \emptyset$, birth threshold S^{c} ;
for k in T **do**
  $\hat{X}(k|k-1) \Leftarrow []$;
  $\hat{X}(k|k) \Leftarrow []$;
  for label l in $\mathbb{L}^{\text{m}} \cup \mathbb{L}^{\text{c}}$ **do**
    $\hat{X}(k|k-1) \Leftarrow [\hat{X}(k|k-1),\ \hat{\mathbf{x}}_l(k|k-1)]$;
  end for
  $\mathbf{A}(k) \Leftarrow$CTDA($\hat{X}(k|k-1)$,$\mathbf{z}(k)$);
  Relevance judging with $\mathbf{A}(k)$;
  for unassociated measurement $\mathbf{z}_u(k)$ **do**
    Generate filter f with initial state using $\mathbf{z}_u(t)$;
    Choosing $l_f \in \mathbb{N}^* \setminus (\mathbb{L}^{\text{m}} \cup \mathbb{L}^{\text{c}})$;
    $\mathbb{L}^{\text{c}} \Leftarrow \mathbb{L}^{\text{c}} \cup \{l_f\}$;
  end for
  for associated pair $(\mathbf{z}_l(k), \hat{\mathbf{x}}_l)_{k|k-1}$ **do**
    Update $\hat{\mathbf{x}}_l(k|k)$ with $\mathbf{z}_l(k)$ using (23) and (24);
  end for
  for unassociated state $\hat{\mathbf{x}}_l(k|k-1)$ **do**
    Update $\hat{\mathbf{x}}_l(k|k)$ using one-step propagation ;
  end for
  for label l in $\mathbb{L}^{\text{m}} \cup \mathbb{L}^{\text{c}}$ **do**
    Calculate track score S_l with Eq. (26);
    if $S_l < 0$ **then**
      Delete f_l;
      $\mathbb{L}^{\text{c}} \Leftarrow \mathbb{L}^{\text{c}} \setminus (l \cap \mathbb{L}^{\text{c}})$;
      $\mathbb{L}^{\text{m}} \Leftarrow \mathbb{L}^{\text{m}} \setminus (l \cap \mathbb{L}^{\text{m}})$;
    end if
    if $S_l > S^{\text{c}}$ and $l \in \mathbb{L}^{\text{c}}$ **then**
      $\mathbb{L}^{\text{c}} \Leftarrow \mathbb{L}^{\text{c}} \setminus (l \cap \mathbb{L}^{\text{c}})$;
      $\mathbb{L}^{\text{m}} \Leftarrow \mathbb{L}^{\text{m}} \cup l$
    end if
  end for
  for $l \in \mathbb{L}^{\text{m}}$ **do**
    $\hat{X}(k|k) \Leftarrow [\hat{X}(k|k), \hat{\mathbf{x}}_l(k|k)]$;
  end for
end for

5. Simulation experiments

In this section, we conduct a thorough performance evaluation of our multiple target tracking framework under various challenging conditions. We first assess the framework's performance in a customized 3D scene dataset developed based on positional measurements. Specifically, we analyze the accuracy of the trajectory association, conduct ablation studies to demonstrate the effectiveness of the designed normalization method and the cross-network structure, and present tracking trajectory results for varying numbers of targets. Subsequently, to benchmark the proposed framework against existing methods, we conduct performance comparisons in a 2D environment, as these widely used multiple target tracking algorithms are predominantly validated in such scenarios.

5.1. CTDA-based association performance evaluation

5.1.1. Dataset and evaluation metric

To evaluate the performance of the proposed CTDA-based network for multiple target association, substantial data is required to ensure generalization and prevent overfitting. Due to the challenges in acquiring sufficient real-world trajectory data, we developed a comprehensive multi-target simulation platform capable of generating large-scale synthetic trajectory measurements for network training.

The training data are generated using the dynamic model in (1) and the measurement model in (2) within a 3D environment. Key parameters of the customized multiple target dataset are summarized in Table 2, where $\mathcal{U}(a, b)$ denotes the uniform distribution over interval $[a, b]$, and $\mathcal{N}(\mu, \sigma^2)$ denotes the Gaussian distribution with mean μ and variance σ^2.

In the trajectory generation procedure, we employ the default random number generator (RNG) furnished by NumPy, specifically a 64-bit permuted congruential generator. Throughout the generation process, a consistent default random seed is selected for all random number distributions, encompassing both uniform and Gaussian distributions. A total of 2000 trajectories are generated, with 1800 trajectories used for training and the remaining 200 reserved for testing.

For performance evaluation, we adopt a comprehensive accuracy metric commonly used in target association and tracking tasks. Since one-to-one correspondence between two datasets is often disrupted by false alarms and missed detections, the following metric [48] is employed:

$$\text{Accuracy}(\mathbf{A}, \hat{\mathbf{A}}) = \frac{\sum_{i=1}^{m} \prod_{j=1}^{n} 1(a_{ij}, \hat{a}_{ij}) + \sum_{j=1}^{n} \prod_{i=1}^{m} 1(a_{ij}, \hat{a}_{ij})}{m + n}, \tag{35}$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\hat{\mathbf{A}} \in \mathbb{R}^{m \times n}$ represent the ground-truth association matrix and the association matrix to be evaluated, respectively, a_{ij} and

Table 2
Training dataset parameters.

Parameter	Distribution
Initial velocity range (m/s)	$\mathcal{U}(200, 1500)$
Initial pitch angle range (rad)	$\mathcal{U}(-\pi/2, \pi/2)$
Initial yaw angle range (rad)	$\mathcal{U}(-\pi/2, \pi/2)$
Acceleration range (m/s^2)	$\mathcal{U}(0, 10)$
Target number range	$\mathcal{U}(1, 25)$
Maximum detection range of x, y, z (m)	$[10^6, 10^6, 10^6]^{\mathrm{T}}$
Measurement noise (m)	$\mathcal{N}(0, (1000)^2)$
Missing rate	0.1
Poisson distribution parameter λ	20

Table 3
Ablation studies on normalization methods and cross architectures.

Model	Normalization	Cross architecture	Loss	Accuracy (%)
Transformer	min-max	none	0.0277	91.96
Transformer	Ours	none	0.0158	95.31
CTDA	min-max	row-column & 3-channel	0.0172	94.37
CTDA	Ours	row-column only	0.0093	97.11
CTDA	Ours	3-channel only	0.0112	96.68
CTDA	Ours	row-column & 3-channel	**0.0067**	**98.33**

$\hat{a}_{ij}$ denote the (i, j)-th element of the respective matrices, and

$$1(a_{ij}, \hat{a}_{ij}) = \begin{cases} 1, & \text{when } a_{ij} = \hat{a}_{ij}; \\ 0, & \text{when } a_{ij} \neq \hat{a}_{ij}. \end{cases} \tag{36}$$

5.1.2. Training details

In the proposed cross Transformer network, the input and output data dimensions are set to 40 to ensure dimensional consistency across the network. The Multi-Head Attention (MHA) module utilizes 4 parallel attention heads to enhance feature representation. To mitigate overfitting, a dropout rate of 0.3 is applied for regularization. For association decision-making, an empirical threshold of $\lambda_a = 0.2$ is adopted.

The cross Transformer model was trained for 8 epochs on the training set with a batch size of 256. The Adam optimizer [49] was adopted with a learning rate of 0.0001.

We perform the experiments using Python 3.9 and PyTorch 1.12. Hardware specifications included an Intel i7-8700K processor, 64GB RAM, and an NVIDIA Titan RTX (24GB VRAM), with Ubuntu 20.04 LTS as the operating system. In the proposed cross Transformer network, the input and output data dimensions are set to 40 to ensure dimensional consistency across the network. The Multi-Head Attention (MHA) module utilizes 4 parallel attention heads to enhance feature representation. To mitigate overfitting, a dropout rate of 0.3 is applied for regularization. For association decision-making, an empirical threshold of $\lambda_a = 0.2$ is adopted.

The cross Transformer model was trained for 8 epochs on the training set with a batch size of 256. The Adam optimizer was adopted with a learning rate of 0.0001. In our platform, the total usage of the VRAM is 5292 MB, and the total training time is within 20 minutes.

5.1.3. Ablation study

We evaluate the impact of each improved component through a stepwise ablation study, and the results are shown in Table 3.

As observed, isolating the normalization method enhancement (rows 1 vs. 2) confirmed its effectiveness in significantly lowering loss and boosting accuracy. The architectural upgrade (rows 1 vs. 3) demonstrates that the row-column and 3-channel cross structure meaningfully enhances model performance. Combining both refinements (rows 4-6) proves that jointly applying our normalization technique with the full cross architecture (row-column + 3-channel) delivers optimal results.

5.1.4. Comparison with existing network architectures

To comprehensively evaluate our proposed CTDA framework, we compare its performance against four representative network architectures: FCN, CNN, LSTM (including the LSTM-based DeepDA method [39]), and Transformer [44]. Table 4 presents the quantitative comparison in terms of loss and accuracy metrics.

Table 4
Performance comparison of network architectures.

Method	Loss	Accuracy (%)
FCN	0.0398	90.77
CNN	0.0219	92.13
LSTM (DeepDA [39])	0.0732	87.33
LSTM	0.0201	93.82
Transformer (BAIT [44])	0.0153	95.28
CTDA	**0.0067**	**98.33**

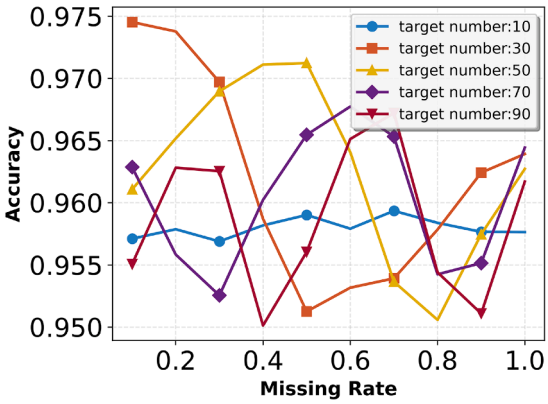


Fig. 5. Accuracy with respect to missing rate.

Table 5
Accuracy with respect to missing rate.

Missing rate	Mean	Variance	Max	Min	Median
0.2	0.963	0.00004	0.974	0.956	0.963
0.4	0.960	0.00005	0.971	0.950	0.959
0.6	0.962	0.00003	0.968	0.953	0.964
0.8	0.955	0.00001	0.958	0.951	0.954
1.0	0.962	0.00001	0.964	0.958	0.963

As shown in Table 4, the FCN architecture achieves moderate accuracy by maintaining spatial relationships through fully-connected layers. However, its global connectivity leads to relatively higher information redundancy, resulting in suboptimal loss values. Both the CNN and standard LSTM demonstrate better performance than FCN, attributed to their specialized architectures for local feature extraction. Notably, our CTDA framework achieves state-of-the-art performance in terms of both accuracy and loss, thanks to the incorporation of multi-scale processing and optimized input normalization. This combination has significantly enhanced the accuracy of data association, which will be exhibited in the subsequent experiments.

5.1.5. Association performance evaluation

We conduct several simulation experiments to evaluate the association performance of the proposed CTDA network under varying conditions, including missing rates, clutter counts, measurement noise levels, and target numbers.

Experiment 1: Accuracy with respect to missing rate

Fig. 5 and Table 5 show the accuracy in relation to the missing rate, with the clutter count fixed at 30 and measurement noise set to zero.

The results indicate that the association accuracy can exceed 95 % when the missing rate is below 0.5. However, accuracy may decrease slightly as the missing rate approaches 1.0, except when the target number is 10. This decline is attributed to a higher relevant false rate that occurs when the target number is low, which adversely affects accuracy.

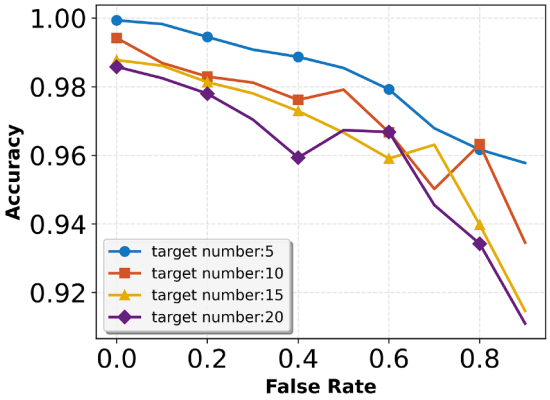


Fig. 6. Accuracy with respect to false rate.

Table 6
Accuracy with respect to false rate.

False rate	Mean	Variance	Max	Min	Median
0.1	0.988	0.00003	0.998	0.983	0.987
0.3	0.980	0.00005	0.991	0.970	0.980
0.5	0.975	0.00006	0.985	0.967	0.973
0.7	0.957	0.00008	0.968	0.946	0.957
0.9	0.930	0.00035	0.958	0.911	0.925

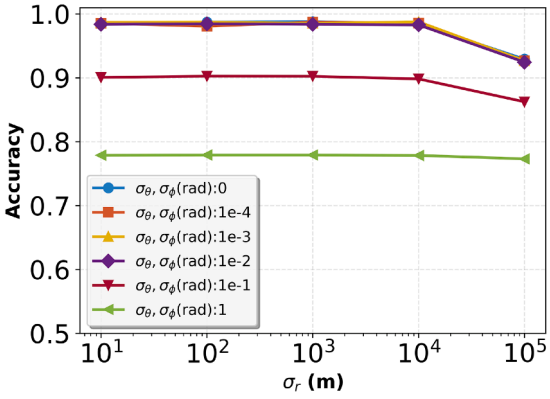


Fig. 7. Accuracy with respect to measurement noise.

Experiment 2: Accuracy with respect to false rate

The false rate varies from 0 % to 100 %. Fig. 6 and Table 6 illustrate the accuracy in relation to the false rates, with noise set to zero.

The result demonstrates that the overall trend in accuracy approaches 96 % as the target number increases. It decreases as the false rate increases, particularly when the target number is 5 or 10.

Experiment 3: Accuracy with respect to measurement noise

We set the false rate and missing rate as 5 % and 0 %, and show the performance of the CTDA method by varying the noise standard deviations σ_θ and σ_r from 0 to 1 rad and 0 to 100 m, as illustrated in Fig. 7.

As seen, the proposed CTDA method achieves an accuracy exceeding 97 % when σ_θ and σ_r are within 0.1 rad and 10 m. Remarkably, it maintains a robust association accuracy of 77 % even under extreme noise conditions (σ_θ and σ_r reaching 1 rad and 100 m), highlighting its resilience to significant noise levels.

Experiment 4: Accuracy with respect to target numbers

We set the number of targets to range from 0 to 100. Fig. 8 and Table 7 illustrate the variation in accuracy relative to the number of targets and the amount of clutter, with measurement noise set to zero.

The result indicates that when the clutter number is low, accuracy generally decreases as the number of targets increases. It is also observed that network performance may decline slightly as the data size

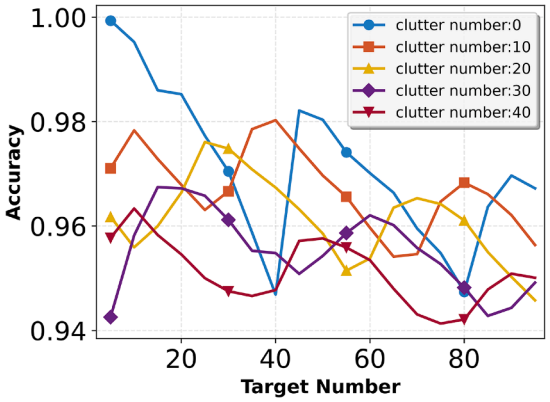


Fig. 8. Accuracy with respect to target numbers.

Table 7
Accuracy with respect to target number.

Target number	Mean	Variance	Max	Min	Median
10	0.970	0.00022	0.995	0.956	0.963
30	0.964	0.00009	0.975	0.948	0.967
50	0.964	0.00009	0.980	0.954	0.959
70	0.956	0.00005	0.965	0.943	0.956
90	0.955	0.00008	0.970	0.944	0.951

approaches 40, corresponding to the padding number. This phenomenon likely arises because the dataset distribution leads to samples near the padding number being underrepresented. While the overall trend for data with different clutter numbers exhibits some fluctuations, the accuracy generally remains stable at about 96 % as the target count increases.

5.1.6. Trajectory tracking results

We further evaluate the performance of our proposed multiple target tracking framework in three-dimensional environments featuring realistic challenges, including measurement noise and missed detections. Fig. 9 presents the comprehensive tracking results from our experimental dataset, where gray markers represent raw measurements and distinct color-coded trajectories represent individual targets for visual distinction.

As observed, the proposed method demonstrates consistent and robust tracking performance across these increasingly complex 3D scenarios. While the system successfully estimates trajectories from noisy measurements and accurately identifies birth/death events, highlighting its advanced capabilities in data association and state estimation, we observe limited instances of trajectory overlap and identity switching in extreme density conditions. These edge cases represent valuable directions for future investigation.

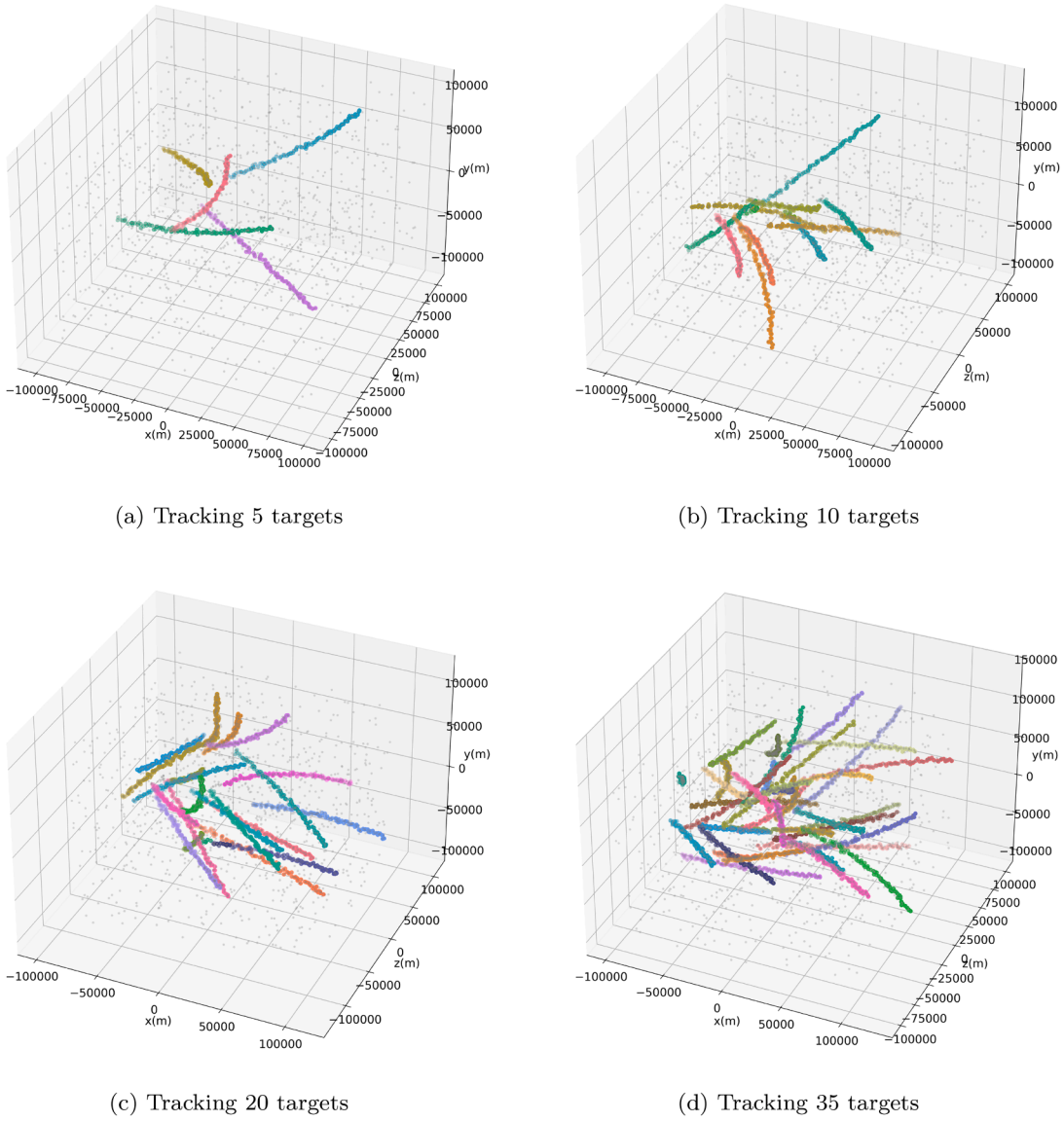
5.2. Overall MTT framework performance evaluation

To evaluate and compare the proposed multiple target tracking algorithm, we conduct comprehensive simulation experiments. Given that random finite set (RFS)-based methods represent the state-of-the-art in target tracking [50], we compare our approach with three RFS-based algorithms, including the probability hypothesis density (PHD) filter [23], the cardinalized PHD (CPHD) filter [51], and the generalized labeled multi-Bernoulli (GLMB) filter [25].

To ensure a fair comparison, all experimental parameters are consistent with the benchmark settings in [50]. The 2D scenario is selected for validation because it represents the standard testing environment for most RFS-based filters.

5.2.1. Dataset and training details

The model used to generate the dataset in 2D environments is aligned with [50]. Let $\mathbf{x}_k = [x, \dot{x}, y, \dot{y}, \omega]^{\mathrm{T}}$ denote the state vector of each



(a) Tracking 5 targets

(b) Tracking 10 targets

(c) Tracking 20 targets

(d) Tracking 35 targets

Fig. 9. 3D multiple target tracking results.

individual target , where $[x, y]$ represents the planar position, $[\dot{x}, \dot{y}]$ denotes the velocity vector and ω is the angular velocity. The discrete-time state equation follows:

$$x_{k+1} = F_\omega x_k + \nu, \tag{37}$$

where

$$F_\omega = \begin{bmatrix} 1 & \frac{\sin\omega}{\omega} & 0 & -\frac{1-\cos\omega}{\omega} & 0 \\ 1 & \cos\omega & 0 & -\sin\omega & 0 \\ 1 & \frac{1-\cos\omega}{\omega} & 0 & \frac{\sin\omega}{\omega} & 0 \\ 1 & \sin\omega & 0 & \cos\omega & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, G = \begin{bmatrix} \frac{1}{2} & 0 \\ 1 & 0 \\ 0 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}, \tag{38}$$

and $\nu \sim \mathcal{N}(0, \Gamma)$ is the process noise with $\Gamma = \mathrm{diag}([\sigma_\omega^2 GG^{\mathrm{T}}, \sigma_u^2])$ with σ_ω and σ_u being the standard deviations.

The measurement equation of noisy detections is given by

$$\mathbf{z}_k = \begin{bmatrix} r \\ \theta \end{bmatrix} + w = \begin{bmatrix} \sqrt{x^2 + y^2} \\ \arctan\frac{y}{x} \end{bmatrix} + w, \tag{39}$$

where r and θ represent the range and the bearing measurement, respectively, and w is assumed to be white measurement noise with σ_r and σ_θ being the noise standard deviations.

Table 8
Parameters of the training dataset.

Parameter	Value Setup
Initial Velocity Range (m/s)	$\mathcal{U}(0, 30)$
Initial Heading Angle Range (rad)	$\mathcal{U}(0, 2\pi)$
Turning Rate Range (rad/s)	$\mathcal{U}(-0.05, 0.05)$
Target Number Range	$\mathcal{U}(1, 25)$
Missing Rate	0.1
Poisson Distribution Parameter λ	20
σ_ω	$15\mathrm{m/s}^2$
σ_u	$\pi/180$ rad/s
R	2000 m
σ_r	5 m
σ_θ	$\pi/360$ rad

The experiment scenarios involve a variable number of targets that evolve through birth, death, and crossing events over time. The complete parameter configuration is detailed in Table 8.

We generate a dataset comprising 2000 trajectories following the specified parameters. Among them, 1800 trajectories are used for the training, while the remaining 200 trajectories are reserved for testing.

Table 9
Monté Carlo simulation result on OSPA distance, location, and cardinality of different methods.

	PHD[23]	CPHD[51]	GLMB[25]	Ours
OSPA distance	17.122	16.430	16.299	**15.148**
OSPA location	**9.691**	11.601	11.752	10.266
OSPA cardinality	7.431	4.829	**4.546**	4.882

The designed cross Transformer network employs a fixed data dimension of 40, matching the dimensionality of both input and output matrices. The architecture incorporates Multi-Head Attention (MHA) modules with 4 parallel attention heads. For regularization, a dropout rate of 0.3 is consistently applied throughout the network. The association threshold parameter λ_a is empirically set to 0.1 for optimal performance.

The cross Transformer network was trained on the training dataset for 8 epochs with a batch size of 256. The Adam optimizer [49] with a learning rate of 10^{-4} was employed.

5.2.2. OSPA distance evaluation metric and results

The Optimal Sub-Pattern Assignment (OSPA) distance [52] serves as a metric for both association accuracy and estimate precision. The OSPA distance quantifies the dissimilarity between two point sets, incorporating both cardinality mismatch and state estimation errors.

Let $P = \{\mathbf{p}_1, \mathbf{p}_2, \cdots \mathbf{p}_m\}$ represent the set of position vectors for real targets, where $\mathbf{p}_i \in P$ denotes the state vector of a real target i, and m indicates the actual number of targets. Correspondingly, let $\hat{P} = \{\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \cdots \hat{\mathbf{p}}_n\}$ be the set of state vectors for estimated target states, where each $\hat{\mathbf{p}} \in \hat{P}$ corresponds to the state estimate of the j-th target and n represents the number of estimates. Without loss of generality, we assume that $m \leq n$. The OSPA distance is defined as:

$$OSPA_r^{(c)}(P, \hat{P}) = \left(\frac{1}{n} \min_{\pi \in \Pi_n} \sum_{i=1}^{m} \left(d^{(c)}\left(\mathbf{p}_i, \hat{\mathbf{p}}_{\pi(i)}\right)^r + (n-m) \cdot c^r \right) \right)^{\frac{1}{r}}, \tag{40}$$

where Π_n denotes the set of all possible injective mappings from P to $\hat{P}$, with cardinality A_n^m, $1 \leq r \leq \inf$ is a distance weight parameter, and $d^{(c)}(\mathbf{p}, \hat{\mathbf{p}}) = \min(c, d(\mathbf{p}, \hat{\mathbf{p}}))$ is the truncated distance between two positions with $c > 0$ being the truncated parameter.

The OSPA distance value can be decomposed into the localization error and the cardinality error. The localization error represents the minimization term that evaluates state estimation accuracy,

$$OSPA_{r,\text{loc}}^{(c)}(P, \hat{P}) = \left(\frac{1}{n} \cdot \min_{\pi \in \Pi_n} \sum_{i=1}^{m} d^{(c)}(\mathbf{p}_i, \hat{\mathbf{p}}_{\pi(i)})^r \right)^{\frac{1}{r}}, \tag{41}$$

and the Cardinality error is

$$OSPA_{r,\text{card}}^{(c)}(P, \hat{P}) = \left(\frac{c^r(n-m)}{n} \right)^{\frac{1}{r}}. \tag{42}$$

where $c^r(n-m)$ penalizes missed detections.

We conduct 100 Monté Carlo simulations on the test dataset. The average OSPA distance over time for each method is presented in Fig. 10 (with $c = 100$ and $r = 1$), while the separate OSPA location and cardinality are illustrated in Figs. 11 and 12, respectively. Table 9 summarizes these results.

As observed, our method achieves tracking accuracy comparable to the PHD while outperforming CPHD and GLMB approaches. Specifically, it exhibits superior OSPA cardinality metrics relative to PHD. As demonstrated in Fig. 12, the proposed approach exhibits rapid adaptation to target birth/death events. Quantitative results in Table 9 show that our method achieves the best overall OSPA distance performance, ranks second in localization accuracy, and maintains competitive cardinality estimation compared to benchmark methods.

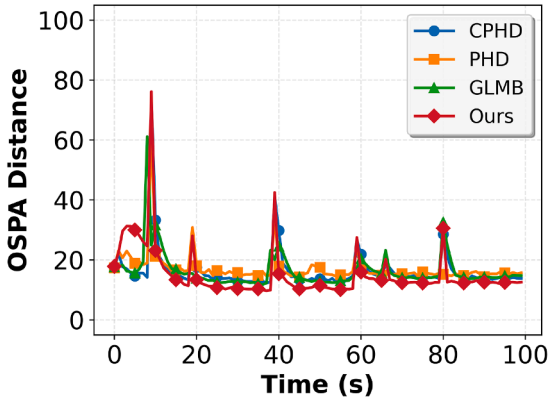


Fig. 10. OSPA distance.

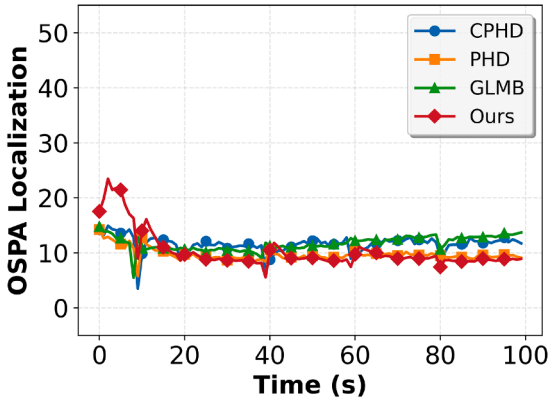


Fig. 11. OSPA location.

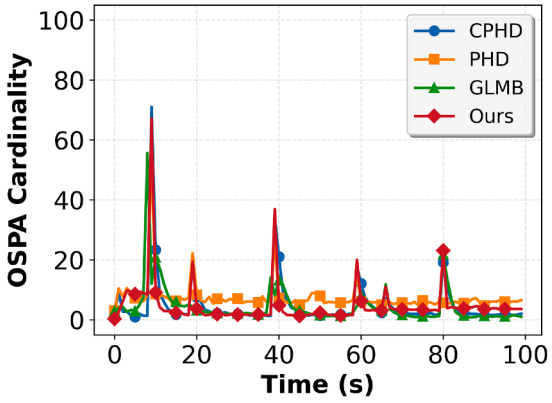


Fig. 12. OSPA cardinality.

5.2.3. Trajectory tracking results

To visualize the performance of the proposed method, we exhibit tracking trajectories in a challenging and representative scenario that involves trajectory crossover, high false alarms, and missed detections. Fig. 13 presents the comprehensive tracking results, while Figs. 14 and 15 respectively illustrate tracking trajectories along x and y directions. In these figures, gray plots represent all measurements (including false alarms), while trajectories corresponding to different targets are distinctively color-coded for clarity.

As seen, the proposed method successfully achieves real-time multiple target tracking while maintaining accurate label assignment for each target. Most importantly, the proposed method exhibits robust recovery capabilities when encountering challenging scenarios, such as temporary target loss due to consecutive false alarms. Furthermore, it ensures

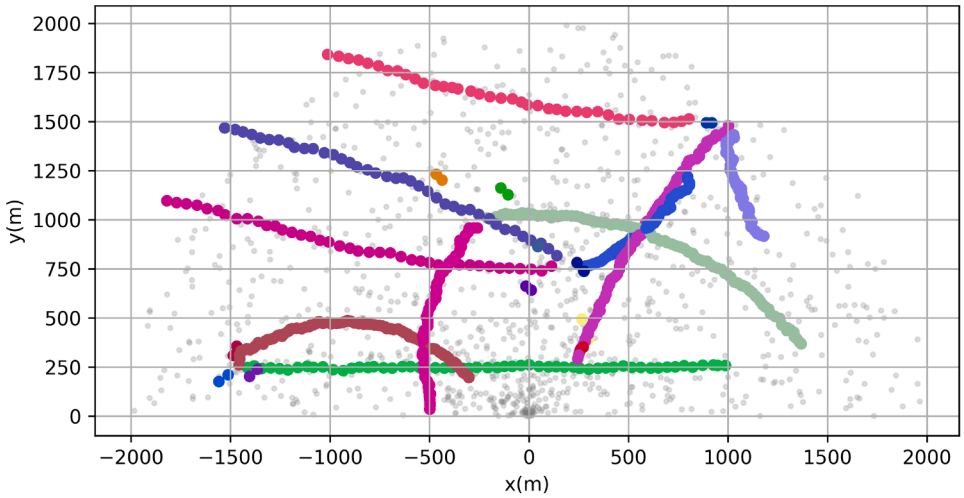


Fig. 13. Tracking trajectories in challenging 2D scenario.

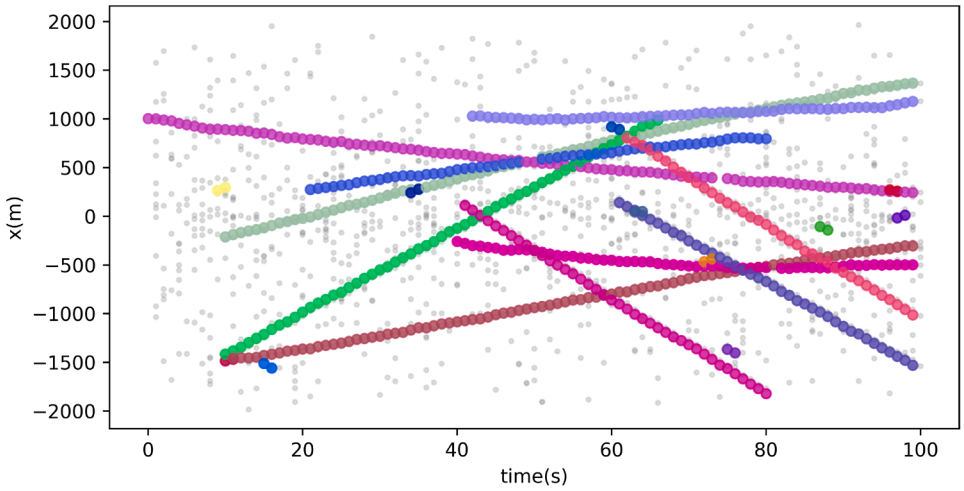


Fig. 14. Tracking result along x direction.

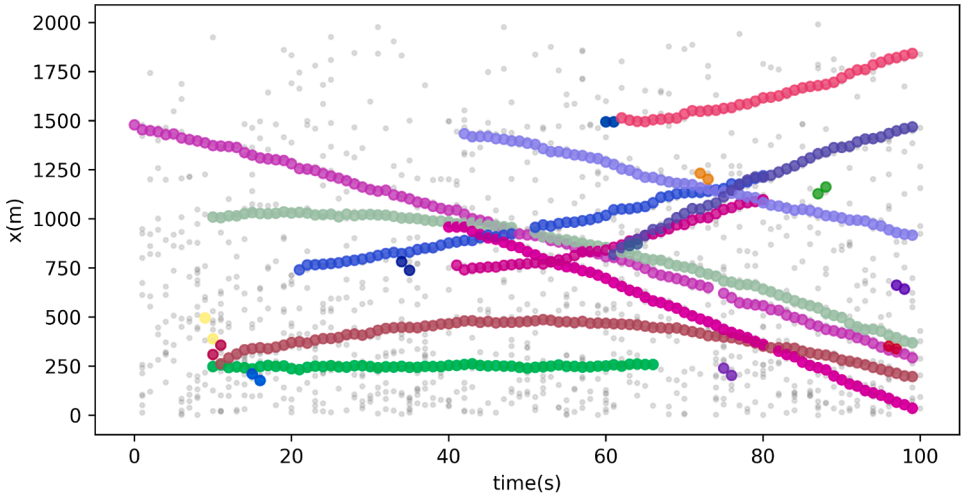


Fig. 15. Tracking result along y direction.

Table 10
Comparison on running time.

Method	Running time (s)
PHD [23]	0.3883
CPHD [51]	0.6653
GLMB [25]	88.37
GLMB (accelerated)	41.62
Ours	**0.2730**

consistent track label maintenance throughout the entire tracking duration without identity switches.

5.2.4. Running time comparison results

To quantitatively evaluate the computational efficiency of our proposed tracking method, we next conduct a comparative analysis of the processing durations. The test settings are described in Section 5.2.1. Table 10 summarizes the total computation time for each approach.

The computational analysis reveals that within the RFS-based framework, the PHD method achieves the highest running speed, followed by the CPHD method. The GLMB, even the accelerated version, suffers from significant computational overhead. Remarkably, our proposed method outperforms all the RFS-based approaches. This is because traditional methods predominantly depend on the computational power of a single CPU core, whereas our deep learning-based CTDA method exploits the parallel processing capabilities of GPUs or multi-core CPUs.

6. Conclusion and future work

In this paper, we investigate the multiple target association and tracking problem using position-only measurements under challenging scenarios involving unknown and varying targets, high missed detections, and false alarms. To tackle these challenging tasks, a deep learning-based association network and an MTT tracking framework are presented. Specifically, we design a Cross Transformer-based Data Association (CTDA) algorithm to effectively extract discriminative features from position measurements. A multiple target tracking framework is proposed in which a trajectory scoring mechanism is introduced to dynamically update each trajectory's confidence score. To achieve reliable tracking, a trajectory management strategy is designed to dynamically handle the full life-cycle of trajectories. Extensive simulation results demonstrate that our proposed method achieves superior performance compared to existing algorithms in terms of accuracy, reliability, and efficiency.

While the proposed method requires less prior knowledge compared to traditional approaches, it still relies on several hyperparameters that need to be configured according to the specific application scenarios. For instance, thresholds for trajectory initiation, association matching, and termination currently require manual tuning. In future work, we plan to replace these hand-crafted settings with learnable mechanisms to enhance adaptability across diverse environments. Additionally, although the proposed method has been evaluated on carefully designed 2D/3D scenarios and established benchmark environments, its generalization capability remains to be further validated across a wider variety of real-world datasets. Future work will include validation on a broader range of real-world datasets to further assess generalization capability.

CRediT authorship contribution statement

Xinran Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Fenghua He:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization; **Yu Yao:** Supervision, Project administration, Funding acquisition, Conceptualization; **Zhaochen Lin:** Writing – review & editing, Validation, Software, Investigation, Data curation.

Data availability

Data will be made available on request.

Declaration of competing interest

All authors disclosed no relevant relationships.

Appendix A. Appendix

Table A.11
Acronyms.

Acronym	Full Name
MTT	Multiple Target Tracking
MOT	Multiple Object Tracking
CTDA	Cross Transformer-based Data Association
BLUE	Best Linear Unbiased Estimation
RFS	Random Finite Set
PHD	Probability Hypothesis Density
CPHD	Cardinalized PHD
GLMB	Generalized Labeled Multi-Bernoulli
LMB	Labeled Multi-Bernoulli
JPDA	Joint Probabilistic Data Association
MHT	Multiple Hypothesis Tracking
PDA	Probabilistic Data Association
NN	Nearest Neighbor
LSTM	Long Short-Term Memory
MHA	Multi-Head Attention
FFN	Feed Forward Network
LN	Layer Normalization
OSPA	Optimal Sub-Pattern Assignment
RMSE	Root Mean Square Error

Table A.12
Important mathematical symbols.

Symbol	Description
$\mathbf{x}_i$	State vector of target i
$\mathbf{z}_j$	Measurement vector j
$\boldsymbol{\eta}_i$	Process noise vector
$\boldsymbol{\nu}_i$	Measurement noise vector
$\mathbf{Q}_i$	Process noise covariance matrix
$\mathbf{R}_i$	Measurement noise covariance matrix
$\mathbf{A}$	Association matrix
$\mathbf{C}^x, \mathbf{C}^y, \mathbf{C}^z$	Relative Distance Feature (RDF) matrices
$\mathbf{O}^{pr}$	Predicted association probability matrix
$\mathbf{O}^{gt}$	Ground truth association matrix
$\mathbf{h}_e, \mathbf{h}_d$	Encoder and decoder hidden features
$S_l(k)$	Confidence score of trajectory l at time k
λ_s	Score decay factor
λ_t	Temporal window size for score truncation
λ_a	Association threshold
P_D	Detection probability
P_{FA}	False alarm probability
P_B	Birth probability
$\mathbb{L}^m$	Set of maintained trajectory labels
$\mathbb{L}^c$	Set of candidate trajectory labels
$\mathcal{N}(\mu, \Sigma)$	Gaussian distribution with mean μ and covariance Σ
$\mathcal{U}(a, b)$	Uniform distribution between a and b
$\mathcal{X}_k$	Set of all target states up to time k
$\mathcal{Z}_k$	Set of all measurements up to time k

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